

Simulation of Galaxy Collision Dynamics using the Toomre Model

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Abstract

When galaxies collide, their gravitational interactions can lead to complex and fascinating structures such as the formation of tidal tails and bridges, as seen in the Antennae Galaxies. These interactions sometimes result in the creation of peculiar features such as tidal tails and shell-like formations in the aftermath of the collision. The Toomre model provides a simple framework for understanding these interactions and their outcomes. In this project, we employed second-order finite difference approximation (FDA) techniques to simulate the Toomre model of close interactions between galaxies. Collisions of co-planar galaxies produced striking tails. Our simulations suggest that the initial velocities and direction of rotation of each galaxy is a significant factor in the tails' structure, with higher initial velocities leading to less pronounced features.



The Antennae Galaxies are a pair of interacting galaxies with dramatic tidal tails formed by their gravitational interaction. Credit: NASA, ESA, and the Hubble Heritage Team (STScI/AURA).

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1 Introduction

The origin and mechanics of large-scale structures in galaxies have been a topic of interest for decades. While some structures are thought to be a result of the formation of their respective galaxies, others, such as "tails" and "bridges," are thought to be the result of interactions between nearby galaxies. With factors such as relative mass, relative orientation, and position, velocity and rotation vectors all influencing the outcome of galactic collisions, myriad large-scale structure types may be produced.

Alar and Juri Toomre model (Toomre 1972) used a simplified model of a galaxy to simulate a range of such interactions. In their model, each galaxy is represented by a core surrounded by a disk of massless stars. The gravitational influence of the core on the star particles and other cores is considered, while the influence of the star particles on each other and on the core is neglected. This simplification allows for simple simulation of complex galactic interactions.

2 Theory

The dynamics of a system of particles under mutual gravitational attraction is governed by Newton's laws of motion and universal gravitation. In this section, we outline the fundamental equations describing the gravitational N -body problem and introduce the Toomre model for simulating galactic collisions.

2.1 The Gravitational N -Body Problem

The gravitational N -body problem describes the motion of N point particles under their mutual Newtonian gravitational interactions. Each particle i has a mass m_i and a position vector

$$\mathbf{r}_i(t) = [x_i(t), y_i(t), z_i(t)]$$

and an acceleration vector

$$\mathbf{a}_i(t) = [a_{xi}(t), a_{yi}(t), a_{zi}(t)]$$

defined in Cartesian coordinates. The force on each particle is the sum of the pairwise gravitational forces from all other particles:

$$m_i \mathbf{a}_i = G \sum_{j=1, j \neq i}^N \frac{m_i m_j}{r_{ij}^2} \hat{\mathbf{r}}_{ij}, \quad (1)$$

where $\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, $r_{ij} = |\mathbf{r}_{ij}|$, and $\hat{\mathbf{r}}_{ij} = \mathbf{r}_{ij}/r_{ij}$ is the unit vector from particle i to j . Using these definitions and dividing through by m_i , we obtain the equations of motion:

$$\frac{d^2 \mathbf{r}_i}{dt^2} = G \sum_{j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}^3}.$$

For convenience, the equations are often non-dimensionalized by choosing units such that $G = 1$, leading to:

$$\mathbf{a}_i = \frac{d^2 \mathbf{r}_i}{dt^2} = \sum_{j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}^3}. \quad (2)$$

To specify a unique solution, initial positions and velocities must be provided:

$$\mathbf{r}_i(0) = \mathbf{r}_{i0}, \quad \mathbf{v}_i(0) = \frac{d\mathbf{r}_i}{dt}(0) = \mathbf{v}_{i0}, \quad i = 1, \dots, N.$$

2.2 The Toomre Model for Galactic Collisions

The Toomre model provides a simplified description of interacting galaxies. Each galaxy consists of a central massive core of mass m_c and a large number of surrounding, massless stars orbiting it. The stars feel the gravitational acceleration of the core(s), but they do not exert forces on other stars or on the cores. In the case of two interacting galaxies, the total acceleration of a given particle is

$$\mathbf{a}_i = \begin{cases} \sum_{j \in \text{cores}, j \neq i} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}^3}, & \text{for cores,} \\ \sum_{j \in \text{cores}} m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{r_{ij}^3}, & \text{for stars.} \end{cases}$$

This structure allows efficient simulation scaling as $\mathcal{O}(N)$ rather than $\mathcal{O}(N^2)$, since only core–star and core–core interactions are computed.

2.3 Star Orbits

Circular orbits around each core are initialized by setting the orbital radii and velocities as

$$v = \sqrt{\frac{m_c}{r}} \quad (3)$$

ensuring centrifugal and gravitational forces balance. The overall dynamics of colliding galaxies are governed by the mutual gravitational attraction between the cores and the perturbation they induce on each other’s stellar distributions.

3 Numerical Approach

This section outlines the numerical strategy used to evolve the Toomre model efficiently and with controlled error. we integrate the equations of motion using an explicit, second-order centered-difference scheme on a uniform time grid. In practice, vectorized kernels are used.

3.1 Discretization

The n-body equations of motions require numerical integration to evolve the system over time. To do that, equation (2) is integrated numerically using a second-order finite difference scheme. Time is discretized uniformly as

$$t_n = (n - 1)\Delta t, \quad n = 1, 2, \dots, n_t,$$

on a uniform grid and continuum domain $0 \leq t \leq t_{\max}$, where $n_t = 2^\ell + 1$ for some integer $\ell \geq 0$. The time step is given by

$$\begin{aligned} \Delta t &= \frac{t_{\max}}{n_t - 1} \\ &= \frac{t_{\max}}{2^\ell} \\ &= t_{\max} 2^{-\ell} \end{aligned}$$

3.2 Centered Finite Difference Scheme

Defining $\mathbf{r}_i^n \equiv \mathbf{r}_i(t_n)$, the second derivative is approximated by the centered finite difference:

$$\left. \frac{d^2 \mathbf{r}_i}{dt^2} \right|_{t=t_n} \approx \frac{\mathbf{r}_i^{n+1} - 2\mathbf{r}_i^n + \mathbf{r}_i^{n-1}}{\Delta t^2}.$$

Substituting this into Eq. (2) yields:

$$\frac{\mathbf{r}_i^{n+1} - 2\mathbf{r}_i^n + \mathbf{r}_i^{n-1}}{\Delta t^2} = \sum_{j \neq i} m_j \frac{\mathbf{r}_j^n - \mathbf{r}_i^n}{r_{ij}^3}.$$

Now, by solving for $\mathbf{r}_i^{n+1}$, we get the discrete update formula:

$$\mathbf{r}_i^{n+1} = 2\mathbf{r}_i^n - \mathbf{r}_i^{n-1} + \Delta t^2 \sum_{j \neq i} m_j \frac{\mathbf{r}_j^n - \mathbf{r}_i^n}{r_{ij}^3} \quad (4)$$

This explicit scheme updates each particle's position at the next time step based on its current and previous positions.

3.3 Initialization

Because the scheme uses three time levels, both $\mathbf{r}_i^1$ and $\mathbf{r}_i^2$ must be known. The first position follows from the initial condition $\mathbf{r}_i(0)$, and the second, $\mathbf{r}_i(\Delta t)$, can be obtained using a Taylor expansion:

$$\mathbf{r}_i^2 = \mathbf{r}_i^1 + \Delta t \mathbf{v}_i(0) + \frac{1}{2} \Delta t^2 \mathbf{a}_i(0),$$

where $\mathbf{a}_i(0)$ is computed from Eq. (2) at $t = 0$, thus:

$$\mathbf{r}_i^2 = \mathbf{r}_i(0) + \Delta t \mathbf{v}_i(0) + \frac{1}{2} \Delta t^2 \sum_{j \neq i} m_j \frac{\mathbf{r}_j(0) - \mathbf{r}_i(0)}{r_{ij}^3} \quad (5)$$

This means that the initial conditions $\mathbf{r}_i(0) = \mathbf{r}_i^1$ and $\mathbf{v}_i(0)$ must be specified for all particles.

4 Implementation

In this section, we describe the implementation details of our numerical solution to the Toomre model using the FDA scheme outlined above.

In our implementation positions for N stars and M cores in a system are stored in 3D arrays `r_stars(N,3,nt)` and `r_cores(M,3,nt)` in MATLAB, respectively. The accelerations are computed at each time step using vectorized operations for efficiency, and the FDA update formula in Eq. (4) is applied iteratively to evolve the system over time.

4.1 N-Body Acceleration

Accelerations are computed by the function `calcToomreAccn(mc,rc,rs)` taking the masses of the cores `mc`, and the positions of the cores `r_cores` and stars `r_stars` and returning `a_cores(M,3)` and `a_stars(N,3)` arrays each of accelerations for the stars and cores, respectively.

4.2 Galaxy Initialization

For galactic collision runs, stars are randomly distributed in around each core within prescribed orbital radii by using the `createGalaxyStars(N,M,R,dim,cwRot)` function in polar coordinates:

$$\begin{aligned} r &= r_{\min} + (r_{\max} - r_{\min}) \cdot \text{rand}(N, 1) \\ \phi &= 2\pi \cdot \text{rand}(N, 1) \\ \theta &= \begin{cases} \pi/2, & \text{(for 2D simulations)} \\ \arccos(1 - 2 \cdot \text{rand}(N, 1)), & \text{(for 3D simulations)} \end{cases} \end{aligned} \quad (6)$$

Stars are given appropriate velocities by Eq. (3) to maintain circular orbits. In 2D, $\mathbf{v}_i^1$ are tangential in the plane, while in 3D they are tangential to the sphere and discussed in detail in section 5.3.3.

4.3 Finite Difference Grid

Time steps are chosen to resolve the shortest dynamical time in the system. This is crucial to accurately capture the rapid changes in acceleration that can occur during close encounters between cores. To preserve the accuracy during each simulation, the shortest star orbital period around any core is calculated as

$$T_{\min} = 2\pi \sqrt{\frac{r_{\min}}{m_c}}$$

The level of refinement ℓ is then chosen such that

$$\Delta t = \frac{T_{\min}}{70} \leq t_{\max} 2^{-\ell} \implies \ell \geq \log_2(t_{\max} \cdot \Delta t)$$

The function `getFiniteDifferenceGrid(tmax,level)` computes the time vector `t` and time step `dt` for a given maximum time `tmax` and refinement level `level`.

4.4 Solving the Trajectory of Particles

The function `calcToomreTrajectory(mc,rc0,vc0,rs0,vs0,dt,nt)` implements the FDA update formula in Eq. (4) in a time-stepping loop to evolve the system. The function takes as input

- `mc` (`M,1`): masses of the cores
- `rc0` (`M,3`): initial positions of the cores
- `vc0` (`M,3`): initial velocities of the cores
- `rs0` (`N,3`): initial positions of the stars
- `vs0` (`N,3`): initial velocities of the stars
- `tmax`: maximum simulation time
- `level`: refinement level for discretization

and initializes the position arrays for stars and cores, setting the first two time steps using the initial conditions and Eq. (5). Then, for each subsequent time step, it computes the accelerations using `calcToomreAccn` and updates the positions using Eq. (4).

4.5 Code Structure Summary

So, our implementation is modular and consists of the following key functions and scripts:

Main Functions:

1. `createGalaxyStars(N,M,R,dim)`: Creates initial positions and velocities for stars in a galaxy.
2. `calcToomreAccn(mc,rc,rs)`: Computes accelerations on cores and stars.
3. `getFiniteDifferenceGrid(tmax,level)`: Generates time grid and step size.
4. `calcToomreTrajectory(mc,rc0,vc0,rs0,vs0,dt,nt)`: Evolves the system using FDA scheme.

Main Scripts:

1. `test2CoresConvergence.m`: Tests convergence with two-body circular orbits.
2. `testSingleGalaxy.m`: Tests the dynamics of a single galaxy.
3. `testDoubleGalaxy.m`: Runs galactic collision simulations for two galaxies.

Auxiliary Functions: (Helpers for Visualization)

1. `createGIF(r_stars, filename, dt, m_cores, r_cores, ndim, lims, freq)`: Visualizes the trajectories of cores and stars.
2. `createStillShot(filename, ics, r_stars, m_cores, r_cores, ndim, simt, lims)`: Save a still shot of a simulation at a time step.

5 Results and Discussion

In this section we presenst the results of our simulations of galactic collisions using the Toomre model and our Implementation of the finite difference approximation method.

5.1 Two Body Case

To verify our implementation, a two-body test case with distinct masses m_1 and m_2 is used. When initialized with the analytic circular orbit parameters,

$$r_1 = \frac{m_2}{m_1 + m_2}r, \quad r_2 = \frac{m_1}{m_1 + m_2}r, \quad v_1 = \sqrt{\frac{m_2 r_1}{r}}, \quad v_2 = \sqrt{\frac{m_1 r_2}{r}} \quad (7)$$

$$\mathbf{r}_1(0) = \begin{pmatrix} r_1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_1(0) = \begin{pmatrix} 0 \\ v_1 \\ 0 \end{pmatrix}, \quad \mathbf{r}_2(0) = \begin{pmatrix} -r_2 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2(0) = \begin{pmatrix} 0 \\ -v_2 \\ 0 \end{pmatrix}$$

Because of the symmetry in the problem, we expect both particles to maintain circular orbits about their center of mass.

5.1.1 Analytic Proof

Consider the two masses with position vectors $\mathbf{r}_1, \mathbf{r}_2$ and $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, $r = |\mathbf{r}|$.

$$m_1 \ddot{\mathbf{r}}_1 = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}, \quad m_2 \ddot{\mathbf{r}}_2 = +G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}},$$

where $\hat{\mathbf{r}} = \mathbf{r}/r$. Subtracting yields the reduced-mass form for the relative motion:

$$\mu \ddot{\mathbf{r}} = -G \frac{m_1 m_2}{r^2} \hat{\mathbf{r}}, \quad \mu = \frac{m_1 m_2}{m_1 + m_2}$$

For a circular orbit $r = |\mathbf{r}|$ is constant and the relative acceleration is purely centripetal $\ddot{\mathbf{r}} = -\omega^2 \mathbf{r}$, so substituting into the reduced equation gives

$$\mu \omega^2 r = G \frac{m_1 m_2}{r^2} \implies \omega^2 = \frac{G(m_1 + m_2)}{r^3}.$$

The distances of each mass from the centre of mass are laid out in Eq. (7) and their tangential speeds follow as $v_i = \omega r_i$. Hence,

$$v_1 = \sqrt{\frac{G m_2^2}{r(m_1 + m_2)}} \quad v_2 = \sqrt{\frac{G m_1^2}{r(m_1 + m_2)}}$$

These expressions show centripetal acceleration $v_i^2/r_i = \omega^2 r_i$ balances the mutual gravitational attraction, proving the circular-orbit solution. (In the unit system, factors of $G = 1$ are omitted.)

5.1.2 Numerical Results

From our simulation result, we confirm our FDA implementation reproduces these circular orbits by simulating two bodies with masses $m_1 = 1$ and $m_2 = 1/2$ and separation $r = 4$.

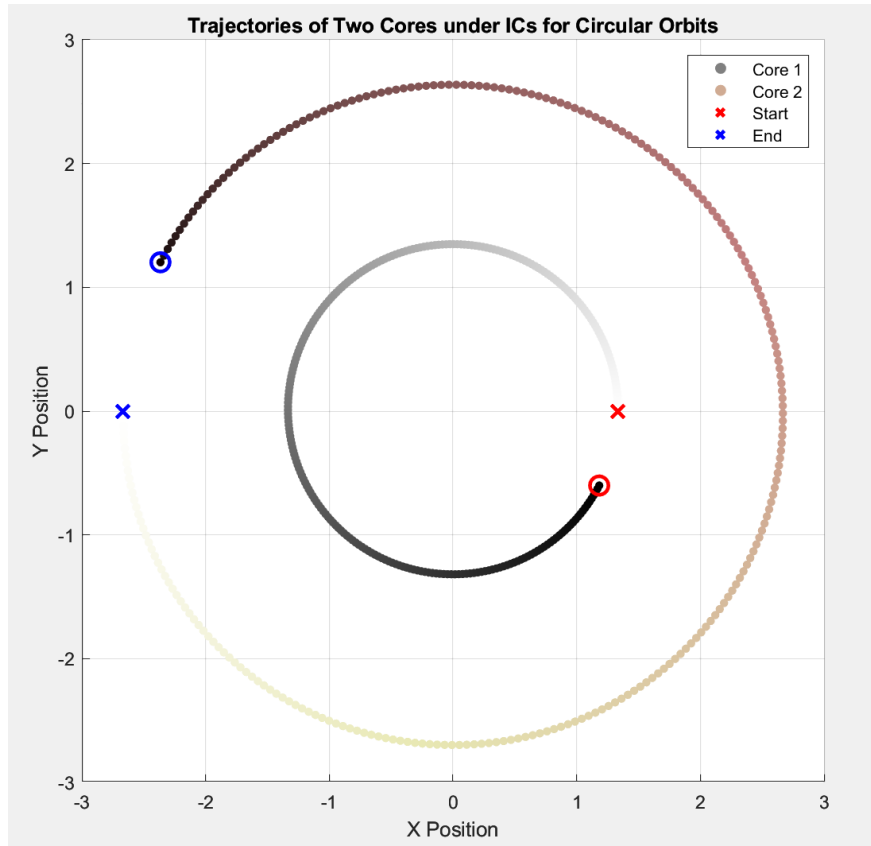


Figure 1: Trajectories of two bodies with masses $m_1 = 1$, $m_2 = 1/2$ and separation $r = 4$ in circular orbits about their center of mass

5.2 Convergence Testing

To verify the second-order accuracy of our numerical solution implementation, we performed a 4-level convergence test for the two-body circular orbit scenario described above. This was done by running simulations with refinement levels $\ell = 6, 7, 8, 9$, corresponding to time steps $\Delta t = t_{\max}/2^\ell$ with $t_{\max} = 38$, and measuring the error in the x -position of one body at the time steps.

We then observe the deviation of the x -positions from each refinement level from each other:

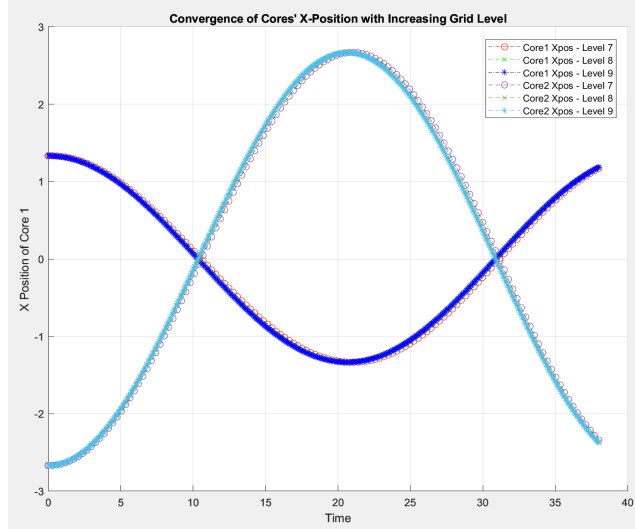


Figure 2: $x_1(t)$ in the two bodies with masses $m_1 = 1$ and $m_2 = 1/2$ with $\mathcal{O}(\Delta t)$ accuracy.

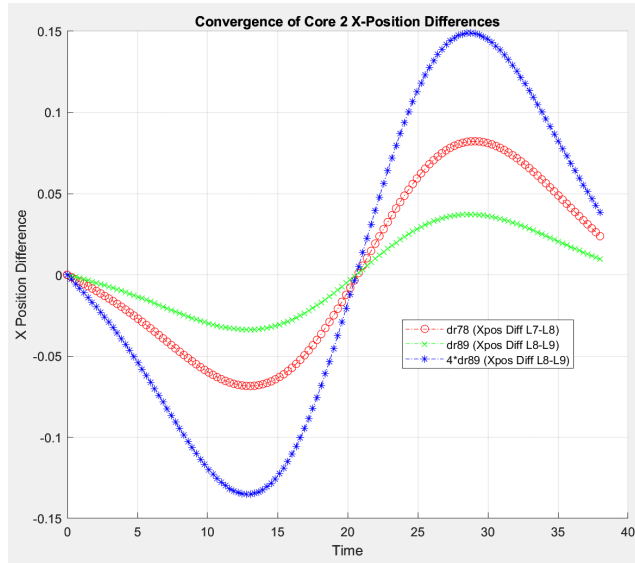


Figure 3: Difference in $x_1(t)$ between refinement levels for the two bodies system with $\mathcal{O}(\Delta t)$ accuracy.

From the results in Figs. 3, we observed that the difference in $x_1(t)$ between refinement levels 9-8 and 8-7 scaled by a factor of 2, which indicated first-order accuracy despite reproducing the circular orbits correctly.

This was traced to an error in the initialization of the second time step causing the scheme to

degrade to first-order accuracy. Initially, we was setting the positions at the second time step without including the acceleration term: $\mathbf{r_stars}(:, :, 2) = \mathbf{rs0} + \mathbf{vs0} * \mathbf{dt}$; This means we was using the Taylor expansion which only provided first-order accuracy:

$$\mathbf{r}(t_0 + \Delta t) = \mathbf{r}(t_0) + \mathbf{v}(t_0)\Delta t + \mathcal{O}(\Delta t^2)$$

For the centered finite difference method requires computing the position at the second time is:

$$\mathbf{r}(t_0 + \Delta t) = \mathbf{r}(t_0) + \mathbf{v}(t_0)\Delta t + \frac{1}{2}\mathbf{a}(t_0)\Delta t^2 + \mathcal{O}(\Delta t^3)$$

With the corrected initialization including the acceleration term, convergence tests confirmed second-order accuracy. When the time step was halved, the positional error decreased by a factor of four, consistent with the expected $\mathcal{O}(\Delta t^2)$ truncation error of the scheme.

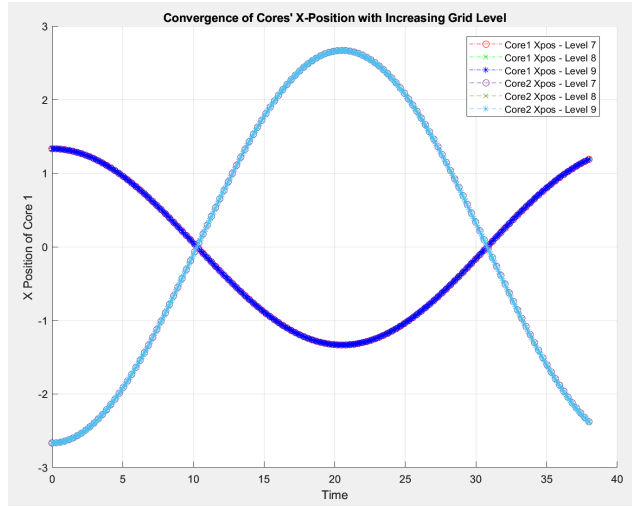


Figure 4: $x_1(t)$ in the two bodies with masses $m_1 = 1$ and $m_2 = 1/2$ with $\mathcal{O}(\Delta t^2)$ accuracy.

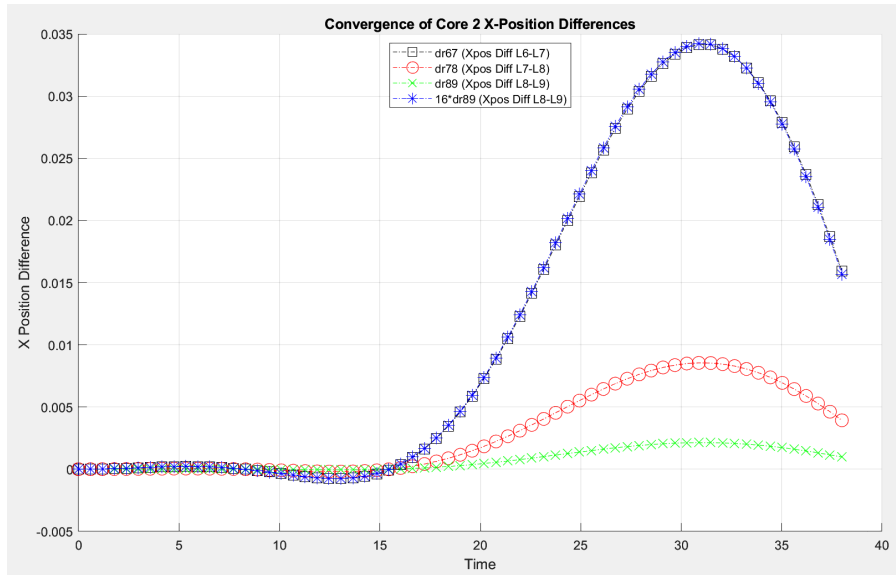


Figure 5: Difference in $x_1(t)$ between refinement levels for the two bodies system with $\mathcal{O}(\Delta t^2)$ accuracy.

Now, in the correct second-order convergences test, Fig. 5, we can see how the magnitude of the absolute error increases with time, which is characteristic of a finite difference solution of this type. The minor deviations observed in Figs. 4 are the expected result of the approximative nature of the finite difference scheme.

Computing the difference between refinement levels 9-8 and 7-6, we observe that the error in $x_1(t)$ scales by the appropriate factor of 16, which is the refinement level difference squared, confirming second-order accuracy. we could continue the convergence test to even higher levels, level ≤ 10 , but we should note that, as discussed in Tutorial 6, for sufficiently high level, the FDA truncation error would [1]. Thus, from the convergence test, we now have strong evidence that the implementation is correct with second-order accuracy.

5.3 Single Galaxy Dynamics

In this section, we validate the dynamics of a single galaxy in isolation to ensure that stars maintain stable orbits around the core as expected.

5.3.1 Creating Stars

To validate the dynamics of a single galaxy in isolation, we simulated a galaxy with a core mass of $m_c = 1$ and 100 stars distributed between orbital radii of $r_{\min} = 0.2$ and $r_{\max} = 2$.

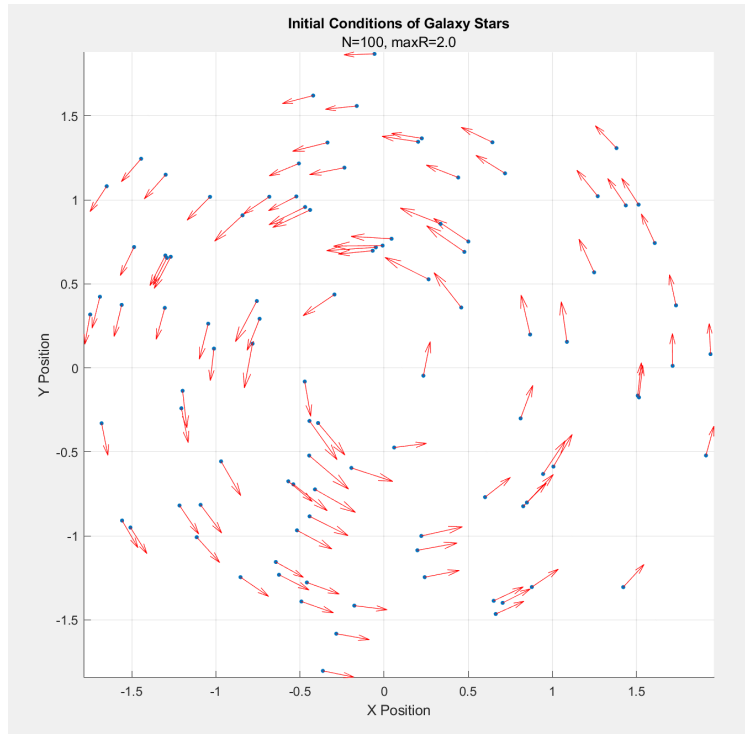


Figure 6: Snapshot of a single galaxy with core mass $m_c = 1$ and 100 stars after evolving for $t = 20$.

In Fig. 6 we verify that `createGalaxyStars()` correctly initializes the stars in circular orbits with tangential velocities around the core as expected from theory.

5.3.2 2D Galaxy Dynamics

Next, we evolved a system with 2000 stars between orbital radii of $r_{\min} = 0.2$ and $r_{\max} = 1$ for $t_{\max} = 10$ with refinement level $\ell = 8$ orbiting around a stationary galaxy with core mass of $m_c = 1$. Fig. 8 is a still shot of the animation and shows that the stars maintain stable circular orbits around the core.

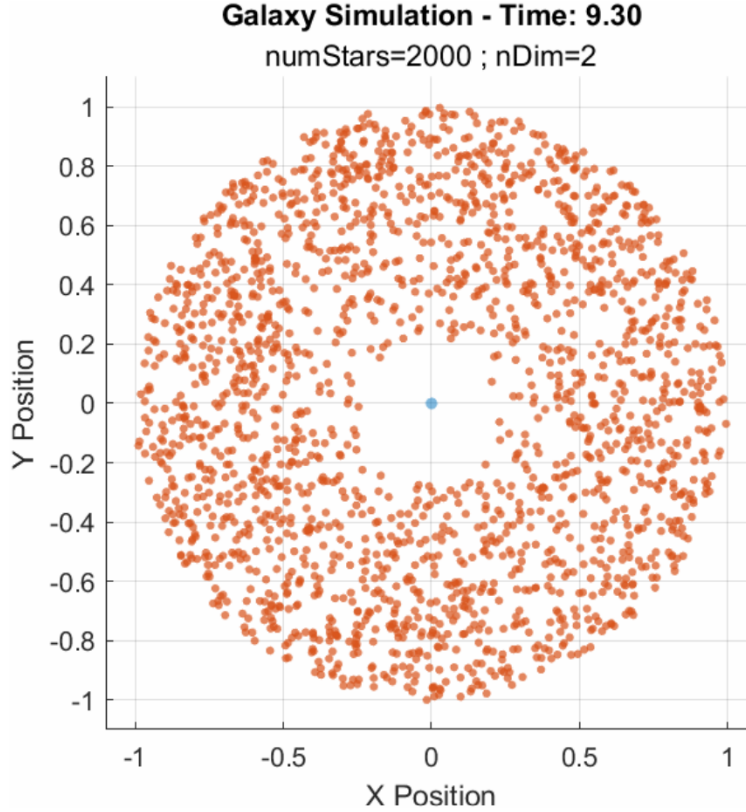


Figure 7: Still shot of 2000 stars evolving around a single galaxy with core mass $m_c = 1$.

The stars remain in stable orbits around the core, confirming that the gravitational acceleration is correctly computed and that the FDA scheme accurately evolves the system over time. GIF of the full simulation was generated using `createGIF()` and available in the supplementary materials folder `./singleGalaxy/singleGalaxyStationary2D.gif`.

Next, we tested a galaxy with an initial translational velocity of $\mathbf{v}_c = (0.5, 0.5, 0)$. To exhibit correct behavior, the galaxy should move uniformly while the stars maintain their orbits around the moving core. This requires correctly adding the core's initial velocity to each star's initial velocity.

$$\mathbf{v}_{\text{star},i}(0) = \mathbf{v}_{\text{orbital},i}(0) + \mathbf{v}_c(0)$$

GIF of the full simulation is available in the at `./singleGalaxy/singleGalaxyMoving2D.gif`.

5.3.3 3D Galaxy Dynamics (Bonus)

We extended the simulation to three dimensions to observe the effects of the additional spatial degree of freedom on the galaxy dynamics. The initial conditions were set up similarly to the 2D case, with stars distributed in a spherical shell around the core. However, now the angle θ in Eq. 6 was randomized to ensure a uniform distribution of stars initial positions, $\mathbf{r}_i^1$, over the sphere.

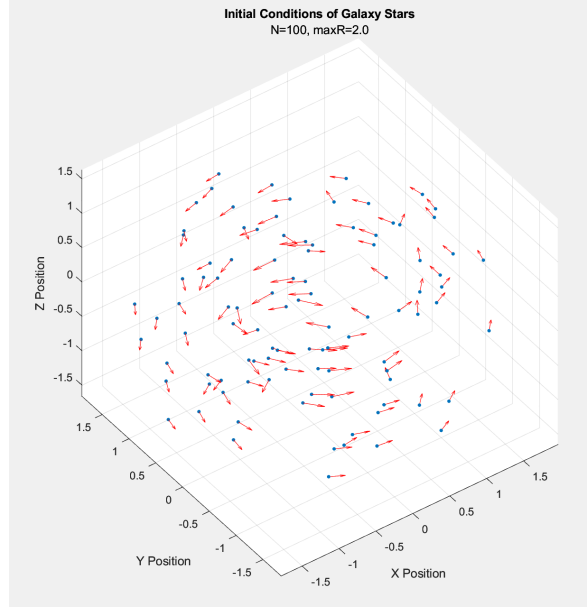


Figure 8: Incorrect initialization of a single galaxy with core mass $m_c = 1$ and 100 stars in 3D.

This resulted in a disk structure rather than a spherical one, as seen in Fig. 9.

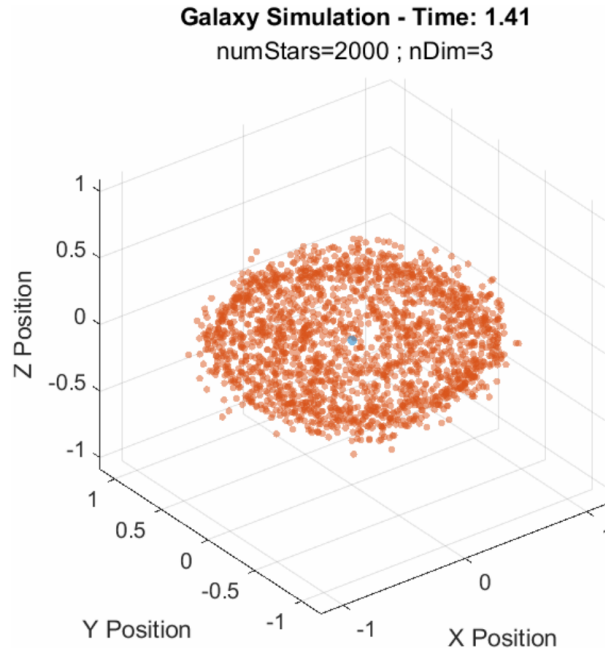


Figure 9: Snapshot of a single galaxy in 3D with incorrect stars velocity initialization.

To correct this, we modified the initialization to ensure that the stars' velocities were set to be tangential to their radial vectors from the core at random orientation. For each star at a random position $\mathbf{r}$ with a random polar angle ϕ , we computed a radial unit vector, and $\mathbf{v}_0$ as follows:

$$\mathbf{u} = \frac{\mathbf{r}}{r}, \quad \mathbf{v}_0 = (-\sin \phi, \cos \phi, 0)^\top.$$

We then added a random tilt angle $\varphi \in (0, 2\pi)$ and in the plane perpendicular to the radial vector and rotated $\mathbf{v}_0$ about the axis $\mathbf{u}$ by angle φ using Rodrigues' rotation formula:

$$\mathbf{v}_{\text{dir}} = \mathbf{v}_0 \cos \varphi + (\mathbf{u} \times \mathbf{v}_0) \sin \varphi + \mathbf{u} (\mathbf{u} \cdot \mathbf{v}_0) (1 - \cos \varphi). \quad (8)$$

So, if the orbital speed is $v_{\text{orb}} = \sqrt{m_c/r}$, the star velocity in 3D is then

$$\mathbf{v}_{\text{star}} = v_{\text{orb}} \mathbf{v}_{\text{dir}}.$$

This adjustment led to a more realistic spherical distribution of stars orbiting the core, as shown in Fig. 10. This galaxy was then given an initial translational velocity of $\mathbf{v}_c = (0.2, 0.2, 0)$ and left to evolve over time.

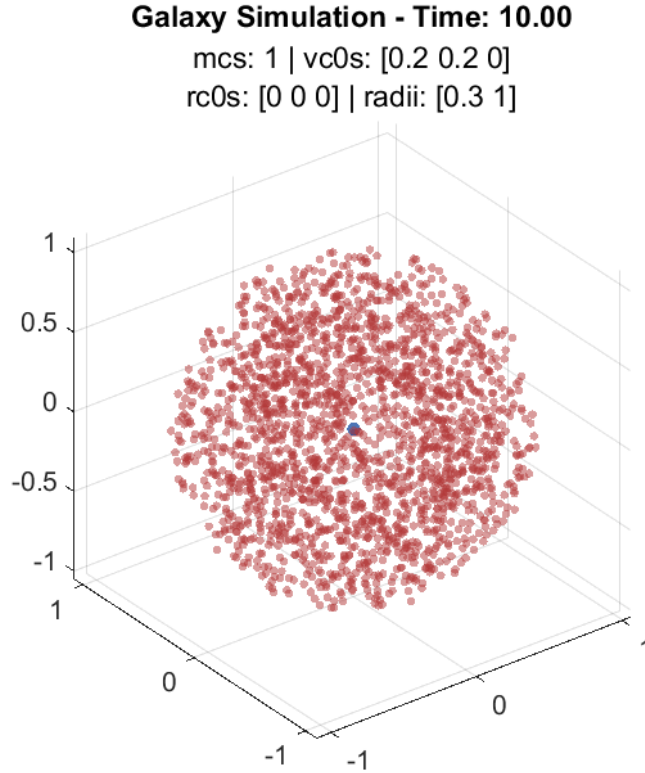


Figure 10: Snapshot of a single galaxy in 3D with truly random initialization.

GIF of the full simulation is available at [./singleGalaxy/singleGalaxyMoving3D.gif](#).

5.4 Double Galaxy Collision Dynamics

Now we proceed to simulate the collision of two galaxies using the Toomre model. There are many parameters that can be adjusted to explore different collision scenarios, including the masses of the galactic cores, the number of stars in each galaxy, and most importantly, their initial positions and velocities. We have chosen some simulations in the following subsections to illustrate the dynamics of galactic collisions.

5.4.1 2D Co-planar Collisions

For co-planar interactions between two galaxies, the significant parameters are the masses, positions, velocities and directions of rotation. With some of these parameters, varying different values for these positions, the simulation was performed to observe the resulting structures formed during the collision.

Grazing Collisions

In grazing collisions, the galaxies are set to pass close to each other without a direct hit, leading to tidal interactions that can produce extended tails and bridges of stars with some "interesting" morphologies, as seen in the Antennae galaxies [1].

The parameters used for these simulations are displayed at the top of each figure with masses, `mcs`, initial positions, `rc0s`, initial velocities, `vc0s`, and orbital radii limits, `radii`.

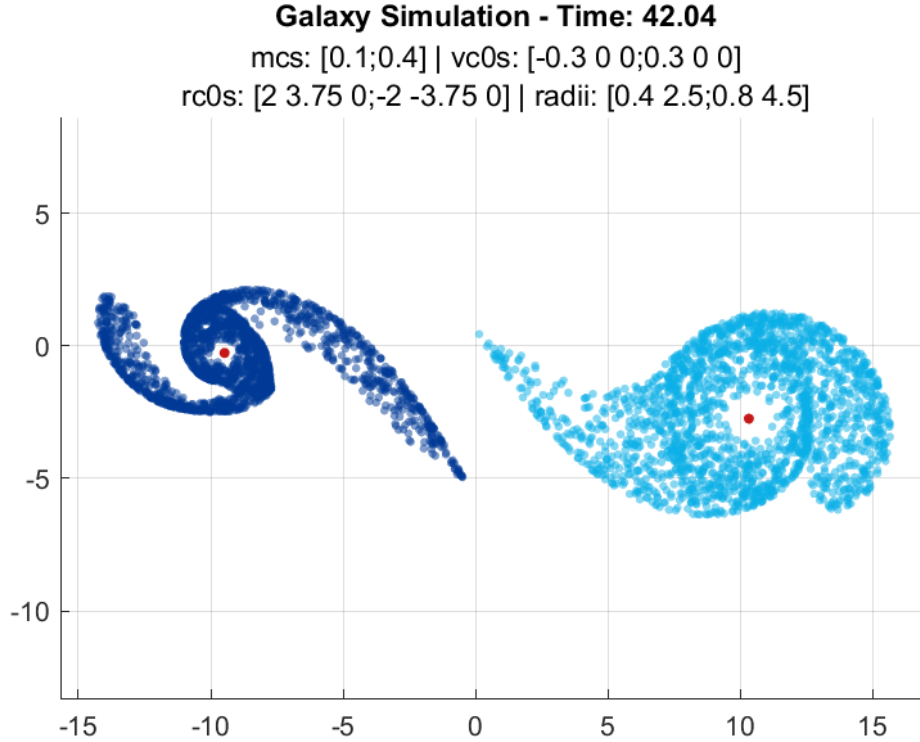


Figure 11: Snapshot of two galaxies undergoing a grazing collision in 2D.

Figures 11 and 12 show snapshots of various grazing collision scenarios.

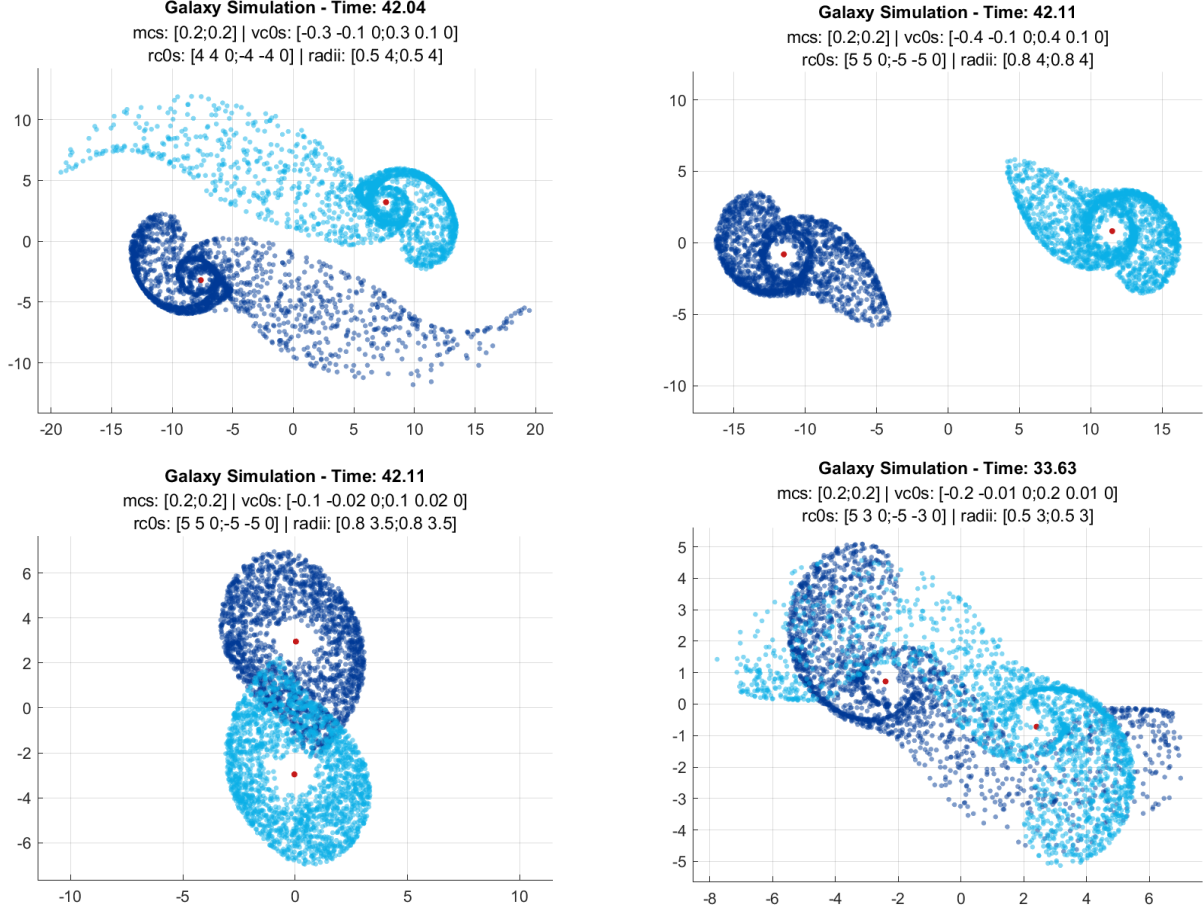


Figure 12: Snapshots of two co-planar galaxies undergoing different grazing collisions in 2D (a) top right, (b) top left, (c) bottom right, (d) bottom left.

From these results, we can conclude that the masses of the cores as well as their initial velocities have significant effects on both the shapes of the tails and in the spaces devoid of stars. Of particular note are the "crossed tails" shown in Fig. 11 which have been elongated asymmetrically compared to the other simulations due to the difference in the masses of the cores, which leads us to conclude that mass ratio plays a significant role in the morphology of the resulting structures.

The significant differences between the structures in each simulation in Figure 12 can be attributed to the separation and the initial velocities of the cores. The galaxies travelling faster past each other like in Fig. 12(b) and have more separation like in Fig. 12(c) tend to produce shorter and less pronounced tidal tails compared to those in Fig. 12(a) and Fig. 12(d) where the galaxies move slower relative to each other. This is because slower encounters allow for stronger gravitational interactions between the galaxies, leading to more significant tidal distortions.

The movies of the full simulations of the grazing collisions in Figs. 11 and 12 are available at

- Fig 11: ./doubleGalaxy/coplanar/doubleGalaxy_1.gif
- Fig 12(a): ./doubleGalaxy/coplanar/doubleGalaxy_2.gif
- Fig 12(b): ./doubleGalaxy/coplanar/doubleGalaxy_3.gif
- Fig 12(c): ./doubleGalaxy/coplanar/doubleGalaxy_4.gif
- Fig 12(d): ./doubleGalaxy/coplanar/doubleGalaxy_5.gif

Perpendicular Collisions

We can also explore a scenario where the galaxies collide travelling perpendicular and towards each other. In such collisions, because the stars and cores get too close to one another and the strong gravitational forces that will not be well-resolved by our FDA scheme, some stars may accelerate uncontrollably. This is why head-on collisions simulation will cease to be meaningful in our FDA framework. Figure. 13 shows a snapshot of a near collision where some stars do get too close to the core and accelerate uncontrollably, but some interesting tidal tails are still formed.

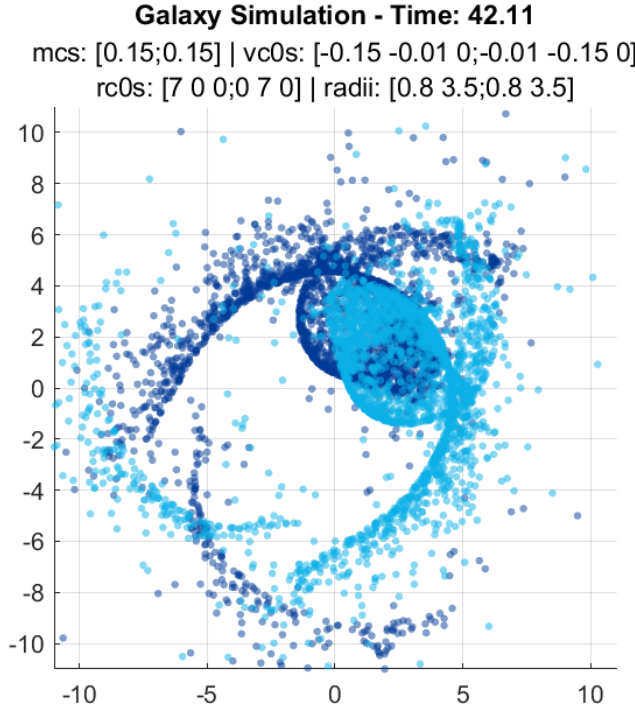


Figure 13: Snapshot of two galaxies travelling perpendicularly in 2D.

5.4.2 3D Parallel Plane Collisions

Another step in investigating galactic interactions is in three dimensions where the galaxy disks' initial conditions are set in parallel planes separated in the vertical direction. We can see that the vertical separation introduces new dynamics not present in the co-planar case, tilting and wrapping the tidal tails vertically as the galaxies interact gravitationally, yet the overall morphology of the tails still resembles those seen in co-planar collisions.

In Figure 14, two galaxy collisions are shown with **in-plane** initial velocities almost identical initial conditions between the two simulation runs, except for the direction of rotation of one galaxy. In Fig. 14(a), both galaxies are rotating clockwise, while in Fig. 14(b), one galaxy is rotating clockwise and the other counter-clockwise. We can see that in the case where the rotations are not aligned, one of the galaxies becomes elliptical and does not form tidal tails like the aligned case. This means that the direction of rotation of each galaxy has a considerable impact on the structures produced during the collision.

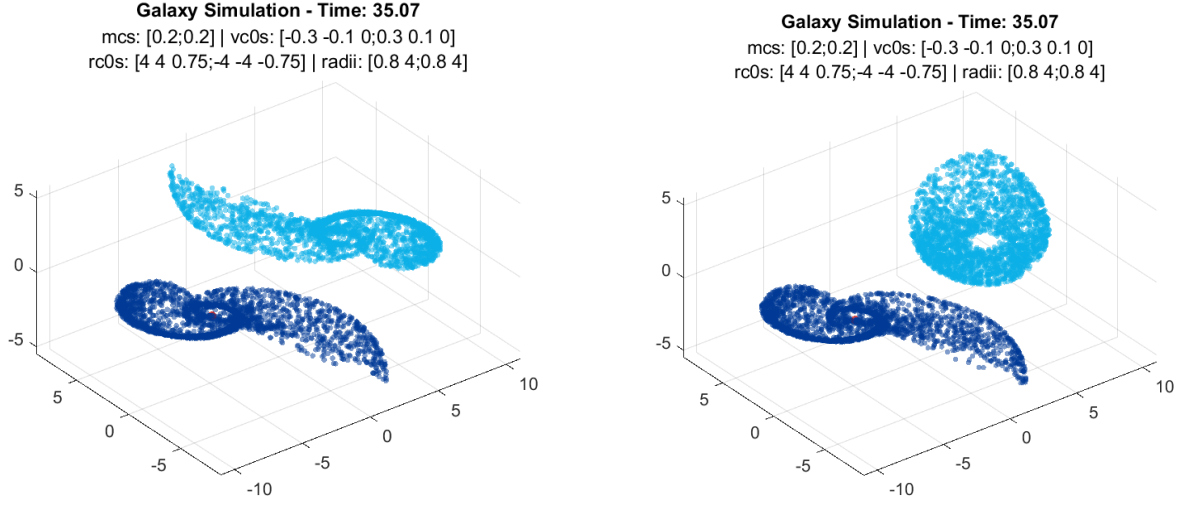


Figure 14: Snapshots of double galaxy collisions in 3D with (a) aligned and (b) anti-aligned rotations.

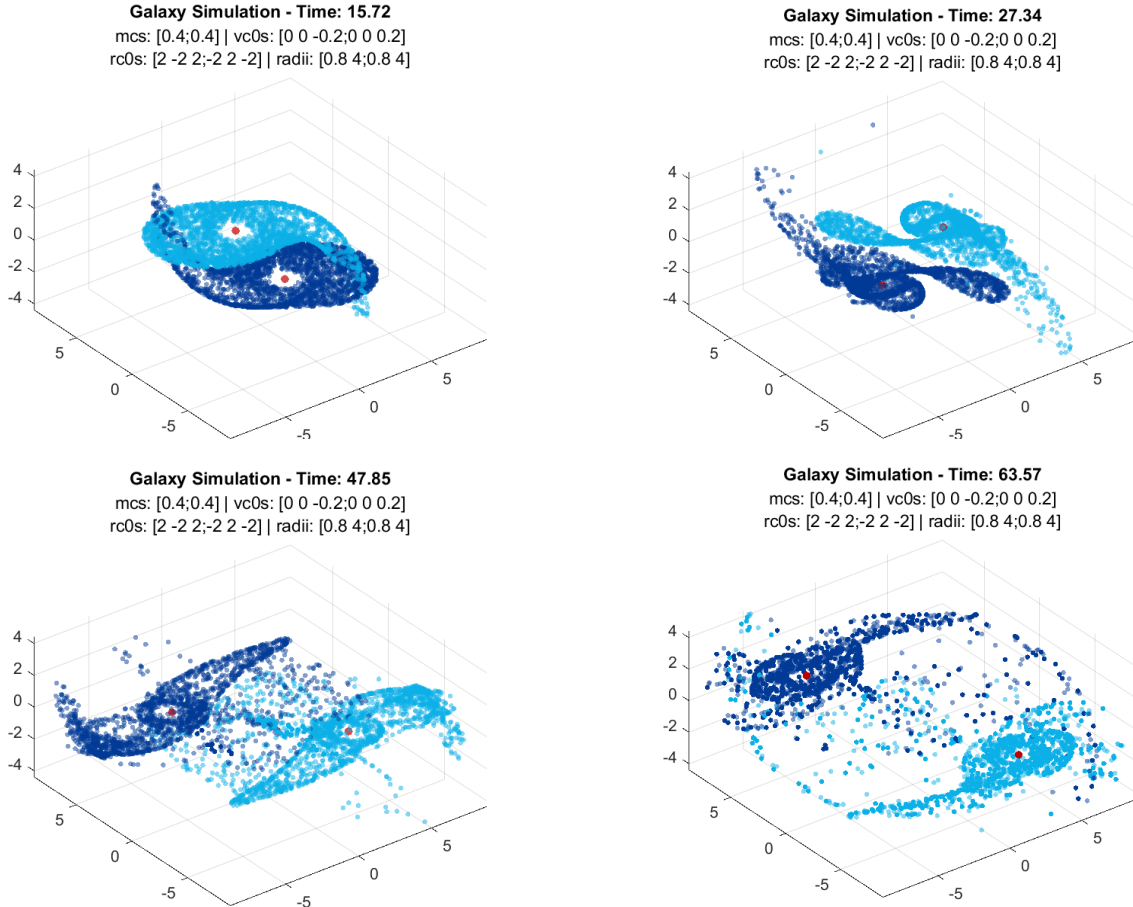


Figure 15: Snapshots of two galaxies travelling vertically towards each other's disk planes in 3D.

In Figure 15, two galaxies collide with **normal to disk plane** initial velocities and snapshots at different times through the simulation are shown.

5.5 Triple Galaxy Collision Dynamics (Bonus)

To further explore the dynamics of galactic interactions, simulating a system of three galaxies was attempted colliding in a co-planar configuration. One result is shown below and more can be found at `./tripleGalaxy/` where similar interaction patterns are observed as in the two-galaxy case.

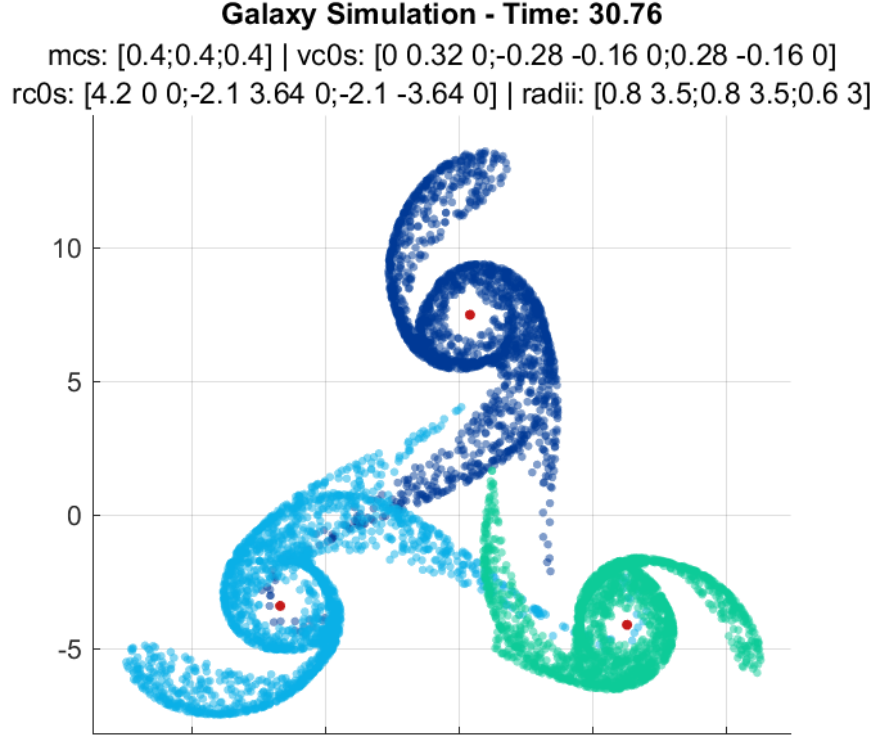


Figure 16: Snapshot of three galaxies undergoing a co-planar collision in 2D.

6 Conclusion

In this project, a second-order finite difference approximation (FDA) scheme was implemented to simulate the dynamics of galactic collisions using the Toomre model. The accuracy of our implementation was verified through convergence testing with a two-body circular orbit scenario and a 4-level grid refinement study, confirming second-order accuracy.

We validated the dynamics of a single galaxy in isolation, ensuring that stars maintained stable orbits around the core. Finally, we explored various collision scenarios between two galaxies in both 2D and 3D configurations, observing the formation of tidal tails and other structures consistent with theoretical expectations. Based on the simulations, we determined that the masses of the galactic cores, their initial velocities, and directions of rotation significantly influence the morphology of the resulting structures formed during the collisions. Aligned rotations and slower travelling galaxies tend to produce more pronounced tidal tails, while anti-aligned rotations can lead to elliptical formations. Heavier cores also result in more asymmetric tail structures, and galaxies separated by larger distances exhibit less interaction patterns due to reduced gravitational influence on each other.

Lastly, a triple galaxy collisions were simulated, revealing similar interactions and the potential for even richer structure formation.

References

- [1] Choptuik, M. (2025). Physics 210: Intro Computational Physics: Course Notes and Resources [Website]. Retrieved from <https://laplace.physics.ubc.ca/410/>
- [2] Toomre, A., and Toomre, J. (1972). Galactic Bridges and Tails. *The Astrophysical Journal*, 178(3), 623-666.